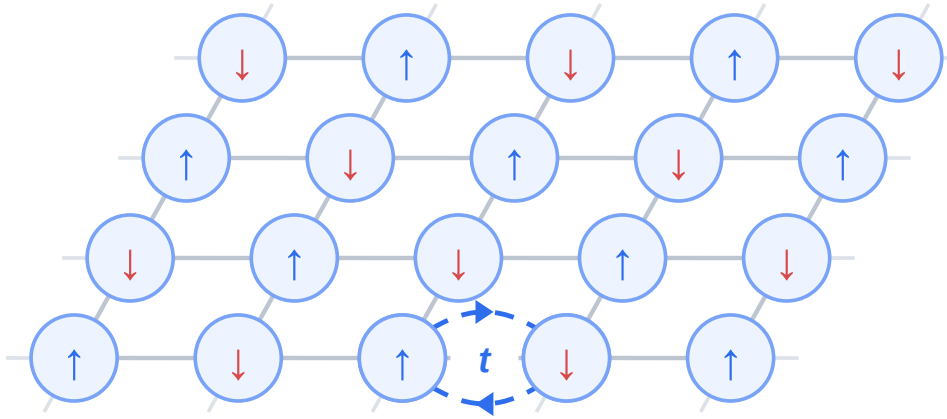
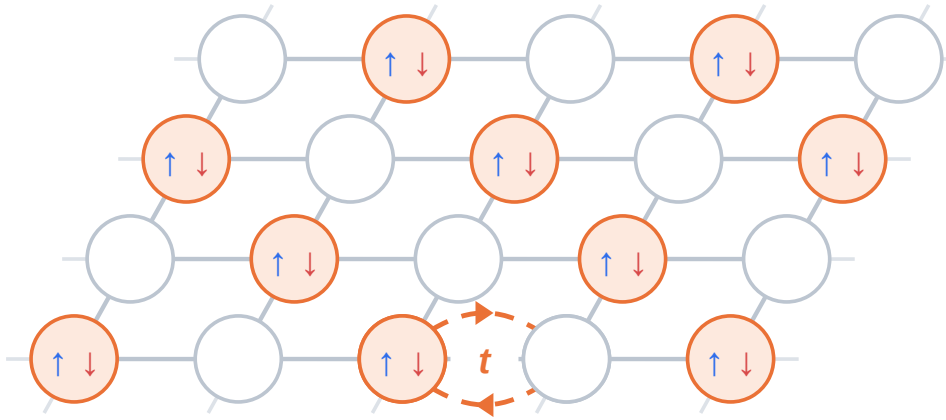
(a) Spin-correlated Mott insulator



$\Delta E_M = U$

(b) Checkerboard CDW insulator



$\Delta E_C = 7V$